

Tamil Offensive Language Detection: Supervised versus Unsupervised Learning Approaches

VIMALA BALAKRISHNAN, VITHYATHERI GOVINDAN, and
KUMANAN N. GOVAICHELVAN, Universiti Malaya

Studies on natural language processing are mainly conducted in English, with very few exploring languages that are under-resourced, including the Dravidian languages. We present a novel work in detecting offensive language using a corpus collected from YouTube containing comments in Tamil. The study specifically aims to compare two machine learning approaches—namely, supervised and unsupervised—to detect offensive patterns in textual communications. In the first setup, offensive language detection models were developed using traditional machine learning algorithms such as Random Forest, Logistic Regression, Support Vector Machine, and AdaBoost, and assessed based on human labeling. Conversely, we used K -means ($K = 2$) to cluster the unlabeled data before training the same set of machine learning algorithms to detect offensive communications. Performance scores indicate unsupervised clustering to be more effective than human labeling with ensemble classifiers achieving an impressive accuracy of 99.70% and 99.87% respectively for balanced and imbalanced datasets, hence showing that the unsupervised approach can be used effectively to detect offensive language in low-resourced languages.

CCS Concepts: • **Computing methodologies** → **Classification and regression trees**; *Ensemble methods*; *Discourse, dialogue and pragmatics*; **Classification and regression trees**;

Additional Key Words and Phrases: Offensive language, Dravidian, Tamil, machine learning, social media

ACM Reference format:

Vimala Balakrishnan, Vithyatheri Govindan, and Kumanan N. Govaichelvan. 2023. Tamil Offensive Language Detection: Supervised versus Unsupervised Learning Approaches. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* 22, 4, Article 106 (March 2023), 14 pages.
<https://doi.org/10.1145/3575860>

1 INTRODUCTION

The proliferation of social media platforms such as Facebook, Instagram, YouTube, and Twitter, among others, has resulted in a colossal amount of user-generated content, with users often using the platforms to seek and share information, establish communications, and vent their feelings. This unfortunately has also opened new avenues for social media users to express their opinions anonymously, thus resulting in the emergence of anti-social behaviors such as cyberbullying, cyberaggression, cyber-trolling, hate speech, misogyny, racism, and abusive language [Alsafari et al. 2020; Vidgen and Derczynski 2020]. The interests in anti-social behavioral studies show an

106

Authors' address: V. Balakrishnan (corresponding author), V. Govindan, and K. N. Govaichelvan, Faculty of Computer Science and Information Technology, Universiti Malaya, 50603 Kuala Lumpur, Federal Territory of Kuala Lumpur, Malaysia; emails: vimala.balakrishnan@um.edu.my, wva180001@siswa.um.edu.my, kumanan@um.edu.my.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2023 Association for Computing Machinery.

2375-4699/2023/03-ART106 \$15.00

<https://doi.org/10.1145/3575860>

incline among not only the researchers but also in natural language processing related competitions, conferences, and workshops. A few examples include HateEval 2019 [Basile et al. 2019], which is a multilingual hate speech detection against immigrants and women, OffensEval 2019 [Zampieri et al. 2019], which focuses on offensive language detection in English, and GermEval 2018 [Wiegand et al. 2018], which targets the German language.

Although there is no formal definition of offensive language, it is often described as speech containing profane, vulgar, or derogatory words (not necessarily abusive in nature) targeting individuals or groups of individuals in ways that are harmful to them Jacobs [2002], Nobata et al. [2016], Alsafari et al. [2020], and Kocon et al. [2021]. Communications that are offensive can create or exacerbate tension, fracture communities, and affect the psychological well-being of affected individuals [Xu et al. 2012; Newton, 2019]. Although lack of empathy and social responsibility have been cited as some of the reasons behind this offensive communication [Ibrohim and Budi 2018], the absence of effective tools to filter offensive language in social media is deemed as a primary reason for this behavior. Considering the large volume of communication taking place online daily, manual monitoring and filtering is not only a laborious task to the human moderators but is almost impossible, hence there is a need for an automated solution, which is made possible with the advent of artificial intelligence. However, offensive language detection is one of the challenging topics for research, as different levels of ambiguities are present in textual communications, and the problem is further exacerbated by the linguistically diverse and varying online communities communicating in various languages laden with dialects, slangs, abbreviations, and so forth Alsafari et al. [2020] and Basile et al. [2019].

Automatic detections are made possible mostly through the supervised machine learning approach (i.e., labeled datasets are required). As a matter of fact, a systematic review analysis of abusive language studies found that most of the studies relied upon human annotators to label the datasets [Vidgen and Derczynski 2020]. Although known to produce more accurate results in classification problems, the supervised approach is time consuming. Further, the authors also revealed that most of the abusive language datasets used have a moderate level of annotation agreement, hence indicating personal biases. Such issues along with a lack of explanation of the ‘experts’ involved in annotation may potentially affect detection performance [Vidgen and Derczynski 2020]. Conversely, the unsupervised approach is based on a clustering mechanism and does not require any labeled datasets. The approach is particularly attractive when there is a tremendous volume of data requiring processing or when languages under study lack resources such as Tamil, Malayalam, and Kannada, collectively known as the Dravidian language.

1.1 The Tamil Language

Dravidian languages mainly represent South Indian languages, comprising 26 languages divided into four branches: 11 in the southern group, 7 in the south-central group, 5 in the central group, and 3 in the northern group [Steever 2019; Chakravarthi et al. 2018]. Spanning across South Asian regions, at least 250 million people can converse in languages belonging to the Dravidian family. Whereas indigenous minority populations use non-literary languages, the South Indian masses use four literary languages—namely Tamil, Malayalam, Kannada, and Telugu—which are used by approximately 210 million people worldwide either as their first or second language [Steever 2019; Chakravarthi et al. 2020b]. This present work focuses on Tamil language, which is deemed to be one of the oldest languages in the world and is used by 68 to 75 million people worldwide with more than 7 million people learning it as a second language [Eberhard et al. 2019]. Therefore, the Tamil language has a worldwide media presence through newspapers, television channels, and websites [Ganesh 2020].

Notably, in social media settings, code mixing among Tamil users has become a norm. As a natural outcome of multilingual use, Tamil people tend to mix languages. Therefore, despite the Tamil language having a native script with Unicode fonts, social media users write Tamil words in Roman alphabets for convenience, which further leads to code mixing [Chakravarthi et al. 2021b]. Examples of code-mixed comments containing offensive and non-offensive languages are given in the following (items in bold are in English):

Example 1: Not offensive

Original: “**Trailer** nalla dhan iruku, theva illama en #BIGIL kooda **clash-u**.”

Translation: *The trailer is good, but why unnecessarily clash with the ‘Bigil’ movie?*

Example 2: Offensive data

Original: “Idhuku **dislike** pota naainga ratham kakki saavuvanunga”

Translation: *Those dogs which disliked this will vomit blood to death.*

1.2 Offensive Language in Tamil

Offensive language refers to any implicit or explicit insult or attack against other people, or any inappropriate language [Haddad et al. 2020]. However, words considered to be offensive can be also used to convey astonishment, amazement, and wonder as noted in some languages, such as the Indonesian language [Ibrohim and Budi 2018]. Therefore, context of the use needs to be taken into consideration. Likewise in Tamil, the type and intensity of the offensive words can be context and community dependent, as listed in the following:

Body parts: This refers to the use of specific body parts as offensive language [Lambert 2014]. For example, in Tamil, the word ‘mayiru’ refers to hair signifying uselessness or something with no value. The overuse of ‘mayiru’ shares grammar similar to the word ‘fuck’ in English, with multipurpose applications [Cusack 2014].

Animal comparison: Some animals are used as offensive or derogatory terms based on the characteristics specific to those animals. For instance, ‘erumai maadu’/‘erume’ (buffalo) either for being big-sized or slow, irresponsible, and inefficient; ‘Naai’/‘Naayi’ (dog) to indicate stupidity or lower standard of living; and ‘Pandi’/‘Panni’ (pig) for being filthy or slow.

Downgrading adjectives: ‘Mokka’ (blunt) or ‘Kuppai’ (rubbish) could be used to refer to something of poor quality. ‘Kirukku’ (lunatic), Keanai (naïve), and ‘Muttaal’ (stupid) are words used to downgrade someone’s cognitive abilities. Some English words have been assimilated into the Tamil community’s colloquial use, such as ‘waste’ (wasteful/ useless), and ‘loosu’ (crazy) to comment on someone’s conduct or action.

Cultural: Certain words or phrases are deemed to be offensive based on the cultural connotation of the community and its living practices [Carson and Jiang 2021]. For example, ‘serupadi’ (slipper shot) is used to humiliate someone (i.e., hitting someone with a slipper is the utmost rudest way of disgracing someone in the Indian community), whereas ‘poramboke’ means unused land or government-owned land, and used to indicate that someone is useless or unworthy.

Social: In India, casteism remains a social issue creating tensions between certain groups of people, hence discussion regarding caste is considered taboo [Kikon 2022]. Thus, words that refer to castes like ‘jaadhi,’ ‘jathi,’ and ‘inam’ can be used to insult a group of people. Words like ‘Parai,’ ‘Paarpaan,’ and ‘Sakkiliyar’ attached with derogatory terms are often used to insult people of a certain class, as they are not socially regarded well.

Religious: Having several religions assimilated among the Tamil community words like ‘moothevi,’ ‘saniyan,’ and ‘saitan’ are used to describe people who bring bad luck or evil omen.

These words are perceived to be negative based on Hinduism and Christianity/Islam-based texts, respectively [Laksana 2021].

1.3 Offensive Language Detection

The extant of the literature revealed English to be the dominant language in offensive language detection studies [Zampieri et al. 2019; Schmidt and Wiegand 2017; Kocon et al. 2021]; however, emerging trends are observed involving non-English studies including Arabic [Alsafari et al. 2020], German [Wiegand et al. 2018], and Hindi [Kumar et al. 2020], among others. This section provides several offensive language detection studies conducted in non-English languages with additional emphasis on those targeting the Dravidian language.

Alsafari et al. [2020] developed a Twitter corpus containing the Gulf Arabic and standard Arabic dialects and compared the performance of machine learning and deep learning algorithms in classifying offensive language. The authors found **Support Vector Machine (SVM)** using word and character embedding to consistently produce high macro-F1 scores compared to the rest of the algorithms. However, Kumar et al. [2018] used **Logistic Regression (LR)** and **Convolutional Neural Networks (CNNs)** to detect aggression using 15,000 Facebook comments in English and Hindi respectively, obtaining a weighted F1-score of 64% for both English and Hindi. An Indonesian study targeting 2,500 tweets using **Naïve Bayes (NB)**, **LR**, **Decision Tree (DT)**, and **Random Forest (RF)** reported the best F-score for NB (i.e., 86%) [Ibrohim and Budi 2018].

As anticipated, we found very few studies targeting the Dravidian languages, with the majority relying on the supervised approach considering the lack of available annotated datasets for the under-resourced languages. The work of Devi et al. [2020] based on the FIRE2015 dataset containing 6,000 Tamil tweets revealed that 5-gram embedding is superior in detecting offensive language compared to trigram embedding using SVM (F1-score: 88%). Sane et al. [2019] analyzed 3,677 tweets in Telugu-English code-mixed language using RF and SVM and further explored the use of character and word N-grams with results indicating SVM to produce the best F-score (i.e., 86%) when combined with hashtags and user mentions. A similar approach was adopted by Chakravarthi et al. [2021a], who detected offensive language among 44,000 Tamil-English code-mixed YouTube comments. Experiments using several machine learning algorithms revealed RF to yield the best results, albeit with a lower F-score of 69%.

Other groups of scholars explored the more advanced deep learning algorithms to detect offensive language. For instance, Anbukkarasi and Varadhaganapathy [2021] used the **Recurrent Neural Network (RNN)**, **Long Short-Term Memory (LSTM)**, Bidirectional LSTM, and **Gated Recurrent Unit (GRU)** to detect offensive language in 225,000 Tamil comments, with results indicating Bidirectional LSTM to produce the highest F1-score (i.e., 95%). A similar study based on transformer (i.e., **Bidirectional Encoder Representations from Transformers (BERT)**) deep learning algorithms using YouTube Tamil and Malayalam comments revealed an F1-score of 90% for the Tamil language [Mandl et al. 2020]. Similarly, Baruah et al. [2021] and Sharif et al. [2021] used a combination of machine learning and BERT variants to detect offensive words using a YouTube Malayalam-Tamil dataset, with best results observed for SVM (F-score = 87%) and XLM-RoBERTA (F-score = 76%) for the Tamil language. Meanwhile, Hande et al. [2021] achieved an F1-score of 74% with YouTube Tamil comments using DistilBERT for offensive language detection. A summary of the studies is presented in Table 1.

Our review revealed no studies exploring the use of unsupervised technique (i.e., using unlabeled datasets) in detecting offensive language in Dravidian languages. To address this gap and to extend the literature on offensive language detection in low-resourced languages, we aim to primarily investigate the differences in using human-labeled datasets and non-labeled datasets with regard to automatically detecting offensive language using a Tamil corpus containing YouTube comments. The next section elaborates on the methodology used in greater detail.

Table 1. Summary of Studies in Offensive Language Detection for Under-Resourced Languages

Reference	Dataset, Language, Size	Technique/Algorithms	Results
Alsafari et al. [2020]	Twitter, Arabic, 5,361	NB, SVM, and LR	SVM with word-char gram: F-score = 85.16%
Kumar et al. [2018]	Facebook, Hindi and English, 15,000	CNN, Multi-layered CNN, LSTM, SVM, multi-task learning (MTL), RF, LR, NB	LR: Hindi F-score = 64% CNN: English F-score = 64%
Ibrohim and Budi [2018]	Twitter, Indonesian, 2,500	NB, RF, SVM	NB with unigram: F-score = 86.43%
Devi et al. [2020]	Twitter, Tamil, 6,000	SVM with N-gram embedding	SVM, 5-gram embedding: F-score = 88%
Anbukkarasi & Varadhaganapathy [2021]	Corpus, Tamil, 225,000	RNN, LSTM, GRU and Bidirectional LSTM with POS Tagging	Bidirectional LSTM: Accuracy = 94% and F-score = 95%
Mandl et al. [2020]	YouTube, Malayalam-Tamil, 64,000	SVM, RNN, LSTM, GRU and Bidirectional LSTM, BERT	SVM (Malayalam), TF-IDF: F-score = 94% BERT (Tamil): F-score = 90%
Chakravarthi et al. [2021a]	YouTube, Tamil-English code mixed, 44,000	LR, SVM, Multinomial NB, K-Nearest Neighbor, DT, RF	RF: F-score = 69%
Sane et al. [2019]	Twitter, Telugu-English code-mixed, 3,677	RF and SVM with Char N-Grams, Word N-Grams, Lexicon, Structural features, Hashtags and Mentions features	SVM with all features: F-score = 86%
Baruah et al. [2021]	YouTube, Malayalam-Tamil, 64,000	SVM and XLM-RoBERTA	SVM (Tamil) (Word features): F-score = 87% XLM-RoBERTA (Malayalam) (Char + Word features): F-score = 97%
Hande et al. [2021]	YouTube, Tamil-English code mixed, 44,000	DistilBERT, ALBERT, RoBERTA, XLM, XLM-RoBERTA, CharacterBERT	DistilBERT (Tamil): F-score = 74%
Sharif et al. [2021]	YouTube, Malayalam-Tamil, 64,000	LR, SVM, LSTM, m-BERT, Indic-BERT, XLM-RoBERTA	XLM-R (Tamil): F-score = 76% XLM-R (Malayalam): F-score = 93%

2 MATERIALS AND METHODS

Figure 1 illustrates the overall methodology used in the present study.

2.1 Dataset Acquisition and Pre-Processing

The dataset used in this study was obtained from a previous study, namely that of Chakravarthi et al. [2021a], who researched several Dravidian languages (i.e., Tamil, Malayalam, and Kannada) mixed with English comments gathered from YouTube. The dataset is publicly available from GitHub.¹ There was a total of 71,701 comments regarding movie trailers in the year 2019. As the

¹<https://github.com/bharathichezhiyan/DravidianCodeMix-Dataset>.

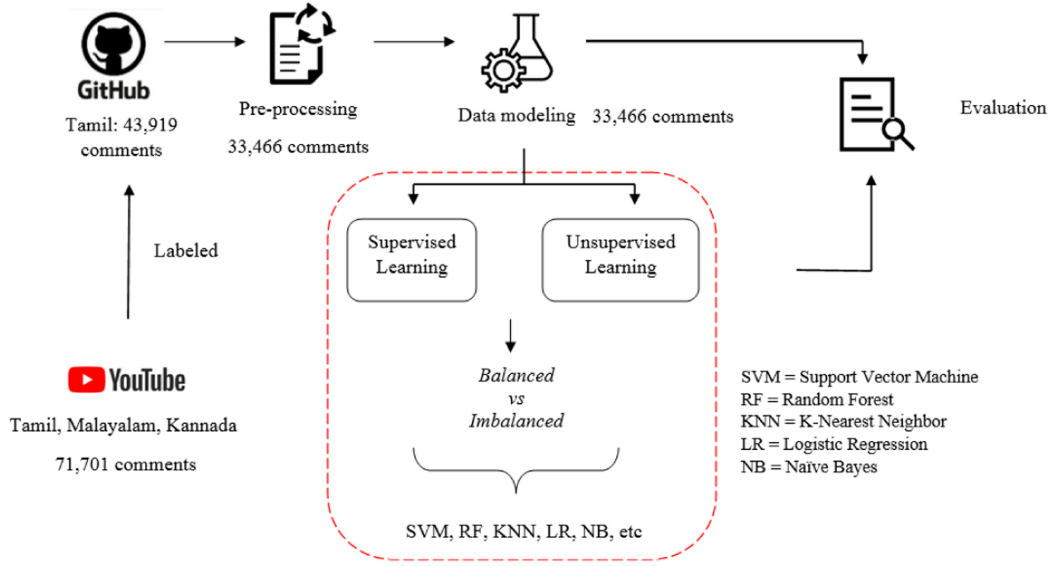


Fig. 1. Overall proposed methodology.

present study is specifically on the Tamil language, only the respective comments were extracted. This resulted in 43,919 Tamil-English comments (i.e., code mixed).

The dataset consists of six labels that were manually annotated by several volunteers, each annotating 100 comments (see Chakravarthi et al. [2021a] for further details). The six labels were Not Offensive, Offensive Untargeted, Offensive Targeted Individual, Offensive Targeted Group, Offensive Targeted Other, and Not in intended language. We further recategorized the labels into two—that is, Offensive and Non-Offensive (i.e., binary classification). In other words, comments labeled as Not Offensive remained the same, whereas all those marked as Offensive were grouped together into a single label. Comments labeled as Not in intended languages and comments of Tamil native script (i.e., Tamil characters) with Unicode fonts were excluded as well. Our final definition is therefore as follows:

Offensive: Comments containing offending words/languages with the intention to hurt or upset someone.

Non-Offensive: Comments not containing any offending words/languages.

The binary classification resulted in a total of 33,466 comments (Offensive = 8,625; Non-Offensive = 24,841). As the dataset is highly imbalanced, appropriate measures were taken to address this issue as elaborated next.

2.2 Dataset: Handling the Imbalanced Issue

To address the skewed distribution of the two classes, the random under-sampling approach was administered in which comments from the majority class (i.e., Non-Offensive) were randomly selected and deleted. This process resulted in a total of 18,625 comments (i.e., Offensive = 8,625; Non-Offensive = 10,000). All modeling and experiments conducted in this study were performed on both the original and the under-sampled datasets for comparison purposes.

2.3 Data Modeling

2.3.1 Supervised Learning Algorithm. SVM, **K-Nearest Neighbor (KNN)**, NB, LR, DT, RF and RF with Bagging, AdaBoost, and Gradient Boosting were the supervised machine learning

algorithms employed in this study. SVM, being the popular algorithm used for text classification, is often hailed for its ability to select the hyperplane to effectively separate binary classes. As selection of kernels influences the SVM's margin of separation, linear kernel is reported to be most suitable for text classification [Kalcheva et al. 2020]. However, unlike SVM, which needs a proper phase of training, KNN classifies data based on distance metrics, which indicates the similarities between data. Thus, selecting the specified number of examples, K , closest to the query is a critical step for the algorithm to calculate the distance between each data point and classify based on the minimal differences. KNN is still widely used, as it is easy to implement and not computationally intensive [Boateng et al. 2020].

Meanwhile, LR and NB are two probabilistic classification algorithms, where LR interprets the maximum likelihood for an event to happen, whereas NB computes the probability of an event to occur based on associated conditions [Das et al. 2020]. However, DT is a rule-based algorithm that incorporates multiple feature nodes to achieve maximum information gain. The decision-making process can be visualized via a tree diagram, whereby a single tree is split into branches based on input features. DT is deemed to be effective for large datasets [Zaman and Hassan 2021].

Conversely, RF, RF with Bagging, AdaBoost, and Gradient Boosting are ensemble learning algorithms that utilize DTs. RF and RF with Bagging construct multiple DTs by splitting the dataset into smaller chunks. The latter algorithm, however, uses the bagging classifier with a base estimator as the RF classifier. Similarly, AdaBoost combines all of the weak learners into a strong learner by assigning the weights through each iteration, whereas Gradient Boosting is similar to AdaBoost but it minimizes loss by merging the weak models [Sarsam et al. 2020; Sindhu et al. 2018; Jain et al. 2017; Bharti et al. 2017]. RandomForestClassifier, AdaBoostClassifier, GradientBoostingClassifier, and BaggingClassifier were imported as part of the Python script implementation from the sklearn.ensemble library. All models were executed using Python with an 80–20 split for training and testing.

2.3.2 Unsupervised Learning Algorithm. Unsupervised machine learning algorithms do not require pre-labeled data to group the data into clusters. The clustering algorithm improves classification performance by grouping these unlabeled data into similar clusters and is considered as one of the less costly alternatives [Jafari-Marandi 2021; Vemulapalli et al. 2015]. The present study adopted the unsupervised algorithm to cluster the unannotated comments (i.e., annotations from the original dataset were excluded) to examine if human labeling and automated labeling play significant roles in offensive language detection performance. Specifically, the simple yet scalable K -means clustering algorithm was used in this study. The algorithm is able to process high-dimensional data [Ramchandran and Sangaiah 2018] and has been widely implemented for large datasets [Mau and Huynh 2021; Kang et al. 2019]. Therefore, it is deemed suitable to use the K -means algorithm, as textual data tend to be highly dimensional with large numbers of features [Zhu 2022].

We used $k = 2$ to group the unlabeled data into two clusters (i.e., Offensive vs Non-Offensive) using the original imbalanced dataset, but with the human labels removed. This process resulted in 28,414 Offensive and 5,052 Non-Offensive comments, and as the outcome is deemed to be highly imbalanced, another round of random under-sampling was performed resulting in a total of 14,346 comments (i.e., Offensive = 9,294; Non-Offensive = 5,052). It is interesting to note that the unsupervised approach yielded a higher number of Offensive labels compared to the human labels (see Section 2.2). Figure 2 illustrates the $k = 2$ clustering for the comments.

It can be observed that the dataset contains data points that are scattered before K -means clustering was applied. As the labeled dataset already consists of two labels, $k = 2$ was selected for grouping the comments into Offensive and Non-Offensive categories. The training point is

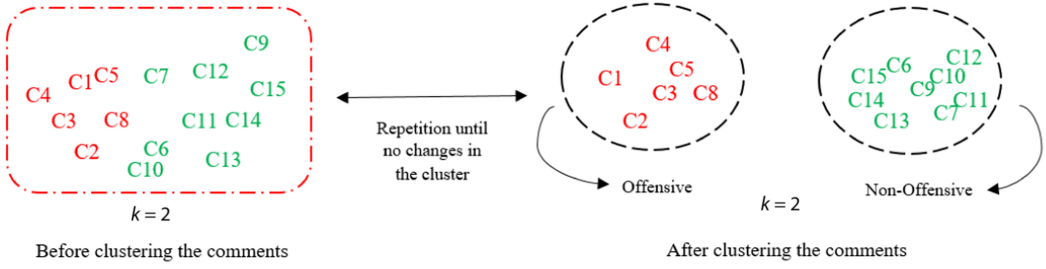


Fig. 2. Overview of k -means clustering ($k = 2$).

inputted into the clusters to determine comments that belong together. Each comment is assigned to their nearest cluster using the Euclidean distance metric. The grouping is based on the smallest distance (i.e., comment that is nearest to the cluster). This step is repeated until no changes are observed in the clusters.

2.4 Experiment and Evaluation

The proposed offensive detection models were assessed in various experimental setups. Specifically, they can be represented using a general convention as given next:

$X_{\text{Imbalanced}}$: Refers to models tested using the original imbalanced dataset, with X referring to each specific algorithm, such as SVM, NB, and LR.

Y_{Balanced} : Refers to models tested using the under-sampled balanced dataset, with Y referring to each specific algorithm, such as SVM, NB, and LR.

Both of the preceding setups were repeated for the unsupervised learning technique as well. A total of 36 models were assessed in completion.

As for the evaluations, the standard performance metrics for classification problems were used to assess both the balanced and imbalanced datasets, namely accuracy, precision (total predicted positive cases), recall (correctly detected positive cases that are actually positive), F-score (balance of precision and recall), and confusion matrix [Alsafari et al. 2020; Jafari-Marandi 2021]. In this study, the micro-average and weighted-average scores were computed for precision, recall, and F-score. The micro-average computes the F-score by considering the total of **True Positives (TP)**, **False Positives (FP)**, and **False Negative (FN)** by giving an average metric, whereas the weighted average returns the average considering the proportion of each label in the dataset [Chakravarthi et al. 2020a, 2021a; Jafari-Marandi 2021].

3 RESULTS AND DISCUSSION

This section discusses the results and performance achieved using both labeling approaches (i.e., human labeling vs unsupervised). A comparison is drawn between the balanced and imbalanced datasets as well.

3.1 Experiment and Evaluation

Table 2 depicts the performance metrics for all supervised learning algorithms evaluated using the original imbalanced and balanced datasets, both containing human labels.

It can be observed that SVM, LR, and NB emerged to be the best detection models for the datasets, with an accuracy ranging between 78% and 83% and F-score ranging between 48% and 81%. The SVM learning model has performed well and achieved the state of the art for text classification, in line with previous studies [Zampieri et al. 2019; Vyshnav et al. 2020], hence indicating that the

Table 2. Supervised Learning Using Human-Labeled Datasets

Model	Algorithm	Accuracy	Micro Average			Weighted Average		
			Precision	Recall	F-score	Precision	Recall	F-score
$X_{\text{Imbalanced}}$	SVM	82.31	0.82	0.82	0.81	0.81	0.82	0.81
	RF	81.21	0.81	0.81	0.81	0.80	0.81	0.79
	RF with Bagging	81.12	0.81	0.81	0.81	0.80	0.81	0.79
	KNN	74.62	0.75	0.75	0.75	0.70	0.75	0.67
	DT	76.46	0.76	0.76	0.76	0.76	0.76	0.76
	LR	82.15	0.82	0.82	0.82	0.80	0.82	0.81
	NB	81.77	0.82	0.82	0.82	0.81	0.82	0.8
	AdaBoost	79.77	0.8	0.8	0.8	0.78	0.8	0.78
	Gradient Boosting	78.29	0.78	0.78	0.78	0.78	0.78	0.73
Y_{Balanced}	SVM	79.09	0.79	0.79	0.79	0.79	0.79	0.79
	RF	76.30	0.76	0.76	0.76	0.76	0.76	0.76
	RF with Bagging	76.21	0.76	0.76	0.76	0.76	0.76	0.76
	KNN	57.05	0.57	0.57	0.57	0.59	0.57	0.48
	DT	71.52	0.72	0.72	0.72	0.72	0.72	0.72
	LR	78.82	0.79	0.79	0.79	0.79	0.79	0.79
	NB	79.09	0.79	0.79	0.79	0.79	0.79	0.79
	AdaBoost	71.25	0.71	0.71	0.71	0.74	0.71	0.71
	Gradient Boosting	72.35	0.72	0.72	0.72	0.73	0.72	0.72

conventional machine learning algorithms are effective in detection offensive language in code-mixed datasets [Alsafari et al. 2020; Kumar et al. 2018; Chakravarthi et al. 2021a].

The comments labeled Offensive and Non-Offensive were correctly predicted by the models. FP and FN indicate the incorrectly predicted Offensive and Non-Offensive comments, and this shows a highly imbalanced ratio of almost 1:3 for the imbalanced dataset as opposed to the balanced dataset, probably due to most prediction models tending to be biased toward the majority class instances [Feng et al. 2021]. The randomly selected comments for the balanced dataset may also have contributed to the fairly balanced ratio of FN and FP for the balanced dataset. As for the imbalanced dataset, this may be a result of comments that do not have any swearing or vulgar intentions, as can be seen in the following example:

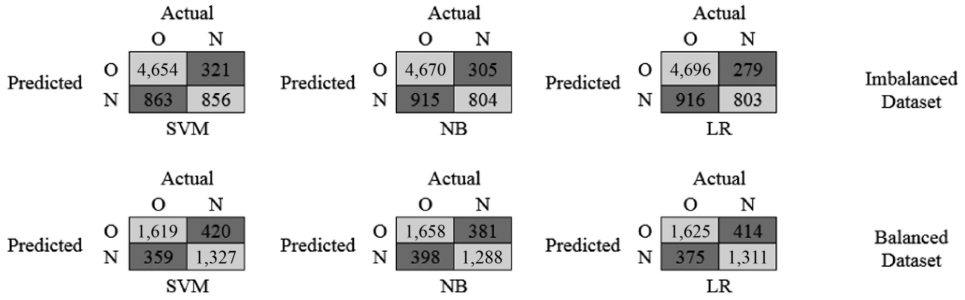
Example 1: *1k + comments in just 12min.*

Example 2: *Vijay and Ajith Anna mutual fan's like mee... I'm telangana... THALLA*

Example 1 is a casual remark on the number of counts that the movie received within a specific time limit, whereas Example 2 is related to a mutual fan for two renowned actors. Both comments do not have any offensive, swearing, or vulgar words; however, in Example 2, it may be labeled as Offensive due to the comment targeting a specific group or race [Chakravarthi et al. 2021a]. Both of the preceding comments were labeled as Offensive in our dataset; however, it is also known that human labeling can be subjective to the annotator's perception of the comment to be offensive [Vidgen and Derczynski 2020].

3.2 Offensive Language Detection Using Clustered Datasets

Table 3 presents the results for the offensive language detection using the datasets that were clustered and labeled using the K -means algorithm. Compared to the human annotated datasets, the overall detection performance is observed to be superior for the clustered datasets, regardless of



Note: O, Offensive; N, Non-Offensive.

Fig. 3. Confusion matrix: human-labeled dataset.

Table 3. Unsupervised Learning Using Clustered Datasets

Model	Algorithm	Accuracy	Micro Average			Weighted Average		
			Precision	Recall	F-score	Precision	Recall	F-score
$X_{\text{Imbalanced}}$	SVM	99.55	1.00	1.00	1.00	1.00	1.00	1.00
	RF	99.87	1.00	1.00	1.00	1.00	1.00	1.00
	RF with Bagging	99.85	1.00	1.00	1.00	1.00	1.00	1.00
	KNN	96.01	0.96	0.96	0.96	0.96	0.96	0.95
	DT	99.75	1.00	1.00	1.00	1.00	1.00	1.00
	LR	99.40	0.99	0.99	0.99	0.99	0.99	0.99
	NB	94.44	0.94	0.94	0.94	0.95	0.94	0.92
	AdaBoost	99.84	1.00	1.00	1.00	1.00	1.00	1.00
Y_{Balanced}	Gradient Boosting	99.85	1.00	1.00	1.00	1.00	1.00	1.00
	SVM	99.52	1.00	1.00	1.00	1.00	1.00	1.00
	RF	99.54	1.00	1.00	1.00	1.00	1.00	1.00
	RF with Bagging	90.87	1.00	1.00	1.00	1.00	1.00	1.00
	KNN	93.26	0.93	0.93	0.93	0.95	0.93	0.94
	DT	99.65	1.00	1.00	1.00	1.00	1.00	1.00
	LR	99.52	1.00	1.00	1.00	1.00	1.00	1.00
	NB	89.29	0.89	0.89	0.91	0.95	0.89	0.91
	AdaBoost	99.70	1.00	1.00	1.00	1.00	1.00	1.00
	Gradient Boosting	99.70	1.00	1.00	1.00	1.00	1.00	1.00

the nature of the datasets (i.e., imbalanced vs balanced). These results indicate that the automated clustering approach can be effectively used to label offensive languages, which is deemed very useful when dealing with low-resourced languages such as the Tamil language.

However, interestingly, the top-performing detection algorithms were found among the ensemble classifiers for both datasets, with accuracies ranging from 90% to 99.87% and F-score ranging between 91% and 100%. It is evident that in an unsupervised learning, ensemble learning performed well by electing some of the best bagging and boosting classifiers, such as for text classification, sentiment analysis, toxic comments [Elnagar et al. 2020; Perikos and Hatzilygeroudis 2016; Haralabopoulos et al. 2020], and medical diagnosis using K-means [Xu et al. 2021]. Figure 3 shows the confusion matrix for the top three models achieving the highest accuracy.

When comparing against the supervised learning labeled dataset, the majority of the Offensive and Non-Offensive labeled data were correctly predicted (TP and **True Negatives (TN)**), as seen in

			Actual					Actual					Actual		Imbalanced Dataset	
			O	N				O	N				O	N		
Predicted	O	415	8	Predicted	O	413	10	Predicted	O	417	6	Predicted	O	417		6
	N	1	6,270		N	0	6,271		N	4	6,267		N	4		6,267
			RF					RF with Bagging					Gradient Boosting			
			Actual					Actual					Actual		Balanced Dataset	
			O	N				O	N				O	N		
Predicted	O	192	10	Predicted	O	192	10	Predicted	O	192	10	Predicted	O	192		10
	N	3	3,520		N	1	3,552		N	1	3,522		N	1		3,522
			DT					AdaBoost					Gradient Boosting			

Note: O, Offensive; N, Non-Offensive.

Note: O, Offensive; N, Non-Offensive.

Fig. 4. Confusion matrix: clustered datasets.

Figure 4. The following examples show the comments that were labeled as Offensive (Example 3) and Non-Offensive (Example 4) during the clustering and the models correctly predicting them:

Example 3: *Ithan padatha matum ranjith pathan vairu erum avanku aiyo unmiya la solitan-galeeee* (If Ranjith watches this movie, he will be upset because the movie telecasts the truth)

Example 4: *Screenplay vera level laaa iruku... Waiting to saw* (The screenplay is amazing... waiting to watch)

Both of the preceding examples were correctly predicted, which corresponds to the TP and TN achieved by the top three models. Example 3 describes the stunned feeling felt by the actor after watching the movie, and Example 4 is just an eager fan waiting for a movie release. For the balanced dataset, FP and FN have a low incorrectly predicted number (less than 12) of Offensive and Non-Offensive comments. For example, in a medical diagnosis, this is an important factor, as an FN is costlier and vulnerable than an FP [Jafari-Marandi 2021] when a diagnosis is not correctly predicted. In the case of offensive language detection, a low FN and FP means that the model has the potential to predict the clustered data correctly based on the available text pattern [Ayo et al. 2020]. This concludes that the clustering conducted using *K*-means contributed to the best model performance. Unsupervised learning performed well compared to the supervised learning, and it can be concluded that unsupervised learning clustering by grouping comments into different clusters helps improve the overall performance and outperforms supervised learning for classification for the Tamil language.

4 CONCLUSION

This study contributed to the evaluation of using human-labeled data compared to those labeled using the unsupervised learning algorithm to detect offensive language using a Tamil-English YouTube dataset containing comments pertaining to South Indian movies. Experiments were conducted using well-known machine learning algorithms, comparing both the imbalanced and balanced datasets. Results generally indicate the traditional machine learning algorithms (i.e., SVM, NB, and LR) to yield the best detection results for the human-labeled datasets (both balanced and imbalanced); however, the ensemble classifiers (i.e., RF, RF with Bagging, and Gradient Boosting) emerged to be the best classifiers for the clustered datasets. It was also found that the unsupervised labeling approach produced superior results compared to the human-labeled dataset, hence showing that clustering can be used effectively to label and detect offensive language for low-resourced languages such as the Tamil language.

REFERENCES

- S. Alsafari, S. Sadaoui, and M. Mouhoub. 2020. Hate and offensive speech detection on Arabic social media. *Online Social Networks and Media* 19 (2020), 100096.
- S. Anbukkarasi and S. Varadhaganapathy. 2021. Deep learning-based Tamil Parts of Speech (POS) tagger. *Bulletin of the Polish Academy of Sciences: Technical Sciences* 69, 6 (2021), e138820–e138820.
- F. E. Ayo, O. Folorunso, F. T. Ibharalu, and I. A. Osinuga. 2020. Machine learning techniques for hate speech classification of Twitter data: State-of-the-art, future challenges and research directions. *Computer Science Review* 38 (2020), 100311.
- A. Baruah, K. A. Das, F. A. Barbhuiya, and K. Dey. 2021. IIITG-ADBU@ HASOC-Dravidian-CodeMix-FIRE2020: Offensive content detection in code-mixed Dravidian text. *arXiv preprint arXiv:2107.14336* (2021).
- V. Basile, C. Bosco, E. Fersini, D. Nozza, V. Patti, F. M. R. Pardo, P. Rosso, and M. Sanguinetti. 2019. SemEval-2019 Task 5: Multilingual detection of hate speech against immigrants and women in Twitter. In *Proceedings of the 13th International Workshop on Semantic Evaluation*. 54–63.
- S. K. Bharti, R. Naidu, and K. S. Babu. 2017. Hyperbolic feature-based sarcasm detection in tweets: A machine learning approach. In *Proceedings of the 2017 14th IEEE India Council International Conference (INDICON'17)*. IEEE, Los Alamitos, CA, 1–6.
- E. Y. Boateng, J. Otoo, and D. A. Abaye. 2020. Basic tenets of classification algorithms k -nearest-neighbor, support vector machine, random forest and neural network: A review. *Journal of Data Analysis and Information Processing* 8, 4 (2020), 341–357.
- L. Carson and N. Jiang. 2021. Collecting and categorizing offensive words in Chinese. In *An Anatomy of Chinese Offensive Words*. Palgrave Macmillan, Cham, Switzerland, 53–65.
- B. R. Chakravarthi, M. Arcan, and J. P. McCrae. 2018. Improving wordnets for under-resourced languages using machine translation. In *Proceedings of the 9th Global Wordnet Conference*. Singapore, 77–86.
- B. R. Chakravarthi, R. Priyadharshini, V. Muralidaran, N. Jose, S. Suryawanshi, E. Sherly, and J. P. McCrae. 2021a. DravidianCodeMix: Sentiment analysis and offensive language identification dataset for Dravidian languages in code-mixed text. *arXiv preprint arXiv:2106.09460* (2021).
- B. R. Chakravarthi, R. Priyadharshini, R. Ponnusamy, P. K. Kumaresan, K. Sampath, D. Thenmozhi, S. Thangasamy, R. Nallathambi, and J. P. McCrae. 2021b. Dataset for identification of homophobia and transphobia in multilingual YouTube comments. *arXiv preprint arXiv:2109.00227* (2021).
- B. R. Chakravarthi, V. Muralidaran, R. Priyadharshini, and J. P. McCrae. 2020a. Corpus creation for sentiment analysis in code-mixed Tamil-English text. *arXiv preprint arXiv:2006.00206* (2020).
- B. R. Chakravarthi, N. Rajasekaran, M. Arcan, K. McGuinness, N. E. O'Connor, and J. P. McCrae. 2020b. Bilingual lexicon induction across orthographically-distinct under-resourced Dravidian languages. In *Proceedings of the 7th Workshop on NLP for Similar Languages, Varieties, and Dialects*. 57–69.
- C. M. Cusack. 2014. Use of the word 'fuck' in pedagogy and higher learning. *Journal of Law & Social Deviance* 8 (2014), 133.
- S. Das, D. Venugopal, and S. Shiva. 2020. A holistic approach for detecting DDoS attacks by using ensemble unsupervised machine learning. In *Proceedings of the Future of Information and Communication Conference*. 721–738.
- G. R. Devi, M. A. Kumar, and K. P. Soman. 2020. Extraction of named entities from social media text in Tamil language using n-gram embedding for disaster management. In *Nature-Inspired Computation in Data Mining and Machine Learning*. Springer, Cham, Switzerland, 207–223.
- D. M. Eberhard, G. F. Simons, and C. D. Fennig. 2019. *Ethnologue: Languages of the World*. SIL International. Available at <https://www.ethnologue.com>.
- A. Elnagar, R. Al-Debsi, and O. Einea. 2020. Arabic text classification using deep learning models. *Information Processing and Management* 57 (2020), 102121.
- S. Feng, J. Keung, X. Yu, Y. Xiao, and M. Zhang. 2021. Investigation on the stability of SMOTE-based oversampling techniques in software defect prediction. *Information and Software Technology* 139 (2021), 106662.
- K. Ganesh. 2020. Complicating 'victimhood' in diaspora studies: The saga of Tamils in exile. *Sociological Bulletin* 69, 3 (2020), 313–330.
- B. Haddad, Z. Orabe, A. Al-Abood, and N. Ghneim. 2020. Arabic offensive language detection with attention-based deep neural networks. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection*. 76–81.
- A. Hande, S. U. Hegde, R. Priyadharshini, R. Ponnusamy, P. K. Kumaresan, S. Thavareesan, and B. R. Chakravarthi. 2021. Benchmarking multi-task learning for sentiment analysis and offensive language identification in under-resourced Dravidian languages. *arXiv preprint arXiv:2108.03867* (2021).
- G. Haralabopoulos, I. Anagnostopoulos, and D. McAuley. 2020. Ensemble deep learning for multilabel binary classification of user-generated content. *Algorithms* 13, 4 (2020), 83.
- M. O. Ibrohim and I. Budi. 2018. A dataset and preliminaries study for abusive language detection in Indonesian social media. *Procedia Computer Science* 135 (2018), 222–229.

- J. B. Jacobs. 2002. Hate crime: Criminal law and identity politics: Author's summary. *Theoretical Criminology* 6, 4 (2002), 481–484.
- R. Jafari-Marandi. 2021. Supervised or unsupervised learning? Investigating the role of pattern recognition assumptions in the success of binary predictive prescriptions. *Neurocomputing* 434 (2021), 165–193.
- T. Jain, N. Agrawal, G. Goyal, and N. Aggrawal. 2017. Sarcasm detection of tweets: A comparative study. In *Proceedings of the 2017 10th International Conference on Contemporary Computing (IC3'17)*. IEEE, Los Alamitos, CA, 1–6.
- N. Kalcheva, M. Karova, and I. Penev. 2020. Comparison of the accuracy of SVM kernel functions in text classification. In *Proceedings of the 2020 International Conference on Biomedical Innovations and Applications (BLI'20)*. IEEE, Los Alamitos, CA, 141–145.
- Z. Kang, H. Xu, B. Wang, H. Zhu, and Z. Xu. 2019. Clustering with similarity preserving. *Neurocomputing* 365 (2019), 211–218.
- D. Kikon. 2022. Dirty food: Racism and casteism in India. *Ethnic and Racial Studies* 45, 2 (2022), 278–297.
- J. Kocon, A. Figas, M. Gruza, D. Puchalska, T. Kajdanowicz, and P. Kazienko. 2021. Offensive, aggressive, and hat speech analysis: From data-centric to human-centered approach. *Information Processing and Management* 58 (2021), 102643. <https://doi.org/10.1016/j.ipm.2021.102643>
- R. Kumar, A. K. Ojha, S. Malmasi, and M. Zampieri. 2018. Benchmarking aggression identification in social media. In *Proceedings of the 1st Workshop on Trolling, Aggression, and Cyberbullying (TRAC-2018)*. 1–11.
- S. S. Kumar, M. A. Kumar, K. P. Soman, and P. Poornachandran. 2020. Dynamic mode-based feature with random mapping for sentiment analysis. In *Intelligent Systems, Technologies, and Applications*. Springer, Singapore, 1–15.
- I. K. D. Laksana. 2021. Discourse of Indonesian language in public domain: Its use in public debate prior to the presidential election 2019. *Linguistics and Culture Review* 5, S1 (2021), 922–934.
- J. Lambert. 2014. Indian English slang. In *Global English Slang*. Routledge, 138–146.
- T. Mandl, S. Modha, M. A. Kumar, and B. R. Chakravarthi. 2020. Overview of the HASOC Track at FIRE 2020: Hate speech and offensive language identification in Tamil, Malayalam, Hindi, English and German. In *Proceedings of the Forum for Information Retrieval Evaluation (FIRE'20)*. 29–32.
- T. N. Mau and V. N. Huynh. 2021. An LSH-based k -representatives clustering method for large categorical data. *Neurocomputing* 463 (2021), 29–44.
- C. Newton. 2019. The trauma floor. *The Verge*. Retrieved November 10, 2021 from <https://www.theverge.com/2019/2/25/18229714/cognizant-facebook-content-moderator-interviews-trauma-working-conditions-arizona>.
- C. Nobata, J. Tetreault, A. Thomas, Y. Mehdad, and Y. Chang. 2016. Abusive language detection in online user content. In *Proceedings of the 25th International Conference on World Wide Web*. 145–153.
- I. Perikos and I. Hatzilygeroudis. 2016. Recognizing emotions in text using ensemble of classifiers. *Engineering Applications of Artificial Intelligence* 51 (2016), 191–201.
- A. Ramchandran and A. K. Sangaiah. 2018. Unsupervised anomaly detection for high dimensional data—An exploratory analysis. In *Computational Intelligence for Multimedia Big Data on the Cloud with Engineering Applications*. Elsevier, 233–251.
- K. R. Sane, S. Kolla, S. R. Sane, V. K. Srirangam, and R. Mamidi. 2019. Corpus and baseline system for hate speech detection in Telugu-English code-mixed tweets. In *Proceedings of the 20th International Conference on Computational Linguistics and Intelligent Text Processing (CICLing'19)*.
- S. M. Sarsam, H. Al-Samarraie, A. I. Alzahrani, and B. Wright. 2020. Sarcasm detection using machine learning algorithms in Twitter: A systematic review. *International Journal of Market Research* 62, 5 (2020), 578–598.
- A. Schmidt and M. Wiegand. 2017. A survey on hate speech detection using natural language processing. In *Proceedings of the 5th International Workshop on Natural Language Processing for Social Media*, Association for Computational Linguistics, Valencia, 1–10. <https://www.aclweb.org/anthology/W17-1101>.
- O. Sharif, E. Hossain, and M. M. Hoque. 2021. NLP-CUET@ DravidianLangTech-EACL2021: Offensive language detection from multilingual code-mixed text using Transformers. *arXiv preprint arXiv:2103.00455* (2021).
- C. Sindhu, G. Vadivu, and M. V. Rao. 2018. A comprehensive study on sarcasm detection techniques in sentiment analysis. *International Journal of Pure and Applied Mathematics* 118, 22 (2018), 433–442.
- S. B. Steever (Ed.). 2019. *The Dravidian Languages*. Routledge.
- S. Vemulapalli, R. J. Dolor, V. Hasselblad, K. Schmit, A. Banks, B. Heidenfelder, M. R. Patel, and W. S. Jones. 2015. Supervised vs unsupervised exercise for intermittent claudication: A systematic review and meta-analysis. *American Heart Journal* 169, 6 (2015), 924–937.
- B. Vidgen and L. Derczynski. 2020. Directions in abusive language training data: Garbage in, garbage out. *arXiv:2004.01670*.
- M. T. Vyshnav, S. Kumar, and K. P. Soman. 2020. Offensive language detection: A comparative analysis. *arXiv preprint arXiv:2001.03131* (2020).
- M. Wiegand, M. Siegel, and J. Ruppenhofer. 2018. Overview of the GermEval 2018 shared task on the identification of offensive language. In *Proceedings of GermEval 2018, 14th Conference on Natural Language Processing*. 1–10.

- J. M. Xu, K. S. Jun, X. Zhu, and A. Bellmore. 2012. Learning from bullying traces in social media. In *Proceedings of the 2012 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. 656–666.
- Z. Xu, D. Shen, T. Nie, Y. Kou, N. Yin, and X. Han. 2021. A cluster-based oversampling algorithm combining SMOTE and *k*-means for imbalanced medical data. *Information Sciences* 572 (2021), 574–589.
- M. Zaman and A. Hassan. 2021. Fuzzy heuristics and decision tree for classification of statistical feature-based control chart patterns. *Symmetry* 13, 1 (2021), 110.
- M. Zampieri, S. Malmasi, P. Nakov, S. Rosenthal, N. Farra, and R. Kumar. 2019. SemEval-2019 Task 6: Identifying and categorizing offensive language in social media (OffensEval). *arXiv preprint arXiv:1903.08983* (2019).
- M. Zhu. 2022. Sentiment analysis of international and foreign Chinese-language texts with multilevel features. *Discrete Dynamics in Nature and Society* 2022 (2022), 1–12.

Received 3 February 2022; revised 12 September 2022; accepted 27 November 2022